Maxwel eqs (等离子体).

$$\nabla \times \vec{E} = \frac{\partial}{\partial t}\vec{B}$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \mu_0 \vec{j} + \frac{1}{c^2}\frac{d\vec{E}}{dt}$$

流体力学

连续性方程.

$$\frac{dN}{dt} = \int_V \frac{dn}{dt} dV = -\oint n\vec{u} \cdot d\vec{s} = -\int \nabla \cdot (n\vec{u}) dV$$

$$\frac{dn}{dt} + \nabla \cdot (n\vec{u}) = 0$$

动量方程:

单粒子

$$nm\frac{d\vec{v}}{dt} = nq(\vec{E} + \vec{v} \times \vec{B}).$$

流体中, 粒子平均速度 = 单个粒子速度. $\vec{u} = 0$.

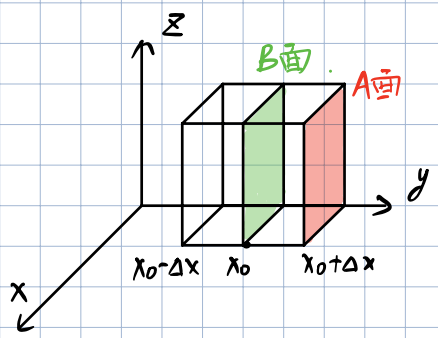
设一维空间中流体 $G(x,t)$.

$$\frac{dG(x,t)}{dt} = \frac{\partial \vec{G}}{\partial t} + \frac{\partial \vec{G}}{\partial x}\frac{\partial x}{\partial t} = \frac{\partial \vec{u}}{\partial t} + u_x \frac{\partial \vec{G}}{\partial x}$$

⇓ 拓展至三维,

$$\frac{d\vec{G}}{dt} + (\vec{u} \cdot \nabla)\vec{G}$$

热运动，



A面，$\Delta n_0 \cdot \vec{v}_x \Delta y \Delta z$.

$$\Delta n_0 = \Delta v_x \iint f(v_x, v_y, v_z)\, dv_y\, dv_z$$

设小立方体中密度为 n，温度 kT.

经A面，

$$P_{A+} = \sum \Delta n_0 \cdot m\vec{v}_x \Delta y \Delta z$$

$$= \Delta y \Delta z \left(m\vec{v}_x^2 \cdot \frac{1}{2} n \right)_{x_0+\Delta x}$$

经B面.

$$P_{B+} = \Delta y \Delta z \left(m\bar{v}_x^2 \cdot \frac{1}{2} n_0 \right)_{x_0}$$

$$P_{A+} - P_{B+} = \Delta y \Delta z \frac{\partial}{\partial x} \left(m\bar{v}_x^2 \cdot \frac{1}{2} n_0 \right).$$

$$\frac{\partial}{\partial t}(nmv_x)\Delta x \Delta y \Delta z = -m_0 \frac{\partial}{\partial x}(n\bar{v}_x^2)\Delta x \Delta y$$

设 $v_x = u_x + v_{xr}$ $\quad \frac{1}{2} m\bar{v}_{xr}^2 = \frac{1}{2} kT$.

$$\therefore \quad \frac{\partial}{\partial t}(nmu_x) = -m_0 \frac{\partial}{\partial x}\left[n\left(\bar{u}_x^2 + 2\overline{u v_{xr}} + v_{xr}^2 \right) \right]$$

$$= -m_0 \frac{\partial}{\partial x}\left[n\left(\bar{u}_x^2 + \frac{1}{2} kT/m \right) \right]$$

$$右 = mn\frac{\partial u_x}{\partial t} + mu_x\frac{\partial n}{\partial t} = -mu_x\frac{\partial (nu_x)}{\partial x} - mnu_x\frac{\partial u_x}{\partial x} - \frac{\partial}{\partial x}(nKT).$$

$$\therefore \quad \frac{\partial n}{\partial t} + \frac{\partial}{\partial x}(nu_x) = 0.$$

Def 圧力 $p \equiv nKT$.

$$mn\left(\frac{\partial u_x}{\partial t} + u_x\frac{\partial u_x}{\partial x}\right) = -\frac{\partial p}{\partial x}.$$

状态方程 $p-n$.

$$p = C\rho^{\gamma} = C\cdot\left(\frac{1}{V}\right)^{\gamma}$$

$$\frac{\nabla p}{p} = \gamma\frac{\nabla n}{n}$$

γ: 自由度 $\frac{C_p}{C_v}$, 等温条件 $\gamma=1$.

$$\nabla p = \nabla(nKT) = KT\nabla n.$$

流体方程.

假定 仅含 离子、电子.

$$\rho = n_i q_i + n_e q_e.$$

$$\vec{J} = n_i q_i \vec{v}_i + n_e q_e \vec{v}_e$$